Yet, remarkably, after a little thought, Mulder realizes that there *is* an experiment that will settle the question completely. Mulder's reasoning is straightforward, but it does require a touch more explicit mathematical reasoning than most things we cover. It's definitely worth trying to follow the details—there aren't that many—but don't worry if some of it slips by; we'll shortly summarize the key conclusion.

Mulder realizes that he and Scully have so far only considered what happens if they each open the same door on a box with a given number. And, as he excitedly tells Scully after calling her back, there is much to be learned if they do not always choose the same door and, instead, randomly and independently choose which door to open on each of their boxes.

"Mulder, please. Just let me enjoy my vacation. What can we possibly learn by doing that?"

"Well, Scully, we can determine whether your explanation is right or wrong."

"Okay, I've got to hear this."

"It's simple," Mulder continues. "If you're right, then here's what I realized: if you and I separately and randomly choose which door to open on a given box and record the color we see flash, then, after doing this for many boxes we must find that we saw the same color flash *more* than 50 percent of the time. But if that isn't the case, if we find that we don't agree on the color for more than 50 percent of the boxes, then you can't be right."

"Really, how is that?" Scully is getting a bit more interested.

"Well," Mulder continues, "here's an example. Assume you're right, and each sphere operates according to a program. Just to be concrete, imagine the program for the sphere in a particular box happens to be *blue, blue, red.* Now since we both choose from among three doors, there are a total of nine possible door combinations that we might select to open for this box. For example, I might choose the top door on my box